

An Analysis of H2B Visa Lotteries

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In this document, I describe how the H2B allocation procedure evolved from first-come first-served to a lottery. I examine the rules of the current lottery, and propose and analyze several alternative designs. I also look into recent data, which suggests that companies are finding loopholes in the application process.

Background

According to the US Department of Labor,

The H-2B nonimmigrant program permits employers to temporarily hire nonimmigrants to perform nonagricultural labor or services in the United States. The employment must be of a temporary nature for a limited period of time such as a one-time occurrence, seasonal need, peakload need or intermittent need.

According to this report based on data from FY2014, landscaping and groundskeeping is by far the largest occupation, accounting for 40% of all H2B visas. Other popular occupations include forest and conservation workers; amusement and recreation attendants; maids and housekeeping cleaners; meat, poultry, and fish cutters; and construction laborers. In fiscal year 2019, 74% of H2B recipients came from Mexico.

In fact, my mom is a designer for a landscaping firm which typically hired about 50 people from Mexico on H2B visas. These were skilled manual laborers who typically came to Minnesota to work each summer, and returned to Mexico to visit family in the winter. But in 2018, her company's request was denied. They scrambled to find people to hire locally, but found that these employees were on the whole less skilled and less professional than those hired on H2B visas. As a result, the company had to significantly scale back its sales goals. Perhaps more importantly, the workers were denied a source of income on which they had previously relied.

This all happened because the government has imposed an annual cap of 66,000 H2B visas.¹ This cap is divided into two allocations of 33,000 visas each: the first for workers who begin employment in the first half of the fiscal year (October 1 - March 31) and the second for workers who begin employment in the second half of the fiscal year (April 1 - September 30). Applications must be submitted no more than 90 days in advance of the proposed start date, so employers who wish to hire workers for the summer months (which is the more popular season) may submit their applications starting on January 1.

Historically, the Office of Foreign Labor Certification (OFLC) processed applications on a first come, first served basis, and treated all applications that arrived on the same day equally. A 2019 report from the Department of Labor outlines how this became a problem:

On January 1, 2018, OFLC received approximately 4,498 applications covering 81,008 worker positions for April 1 start dates of work, exceeding the semi-annual visa allotment by nearly 250 percent.

I guess my mom's company was hardly the only one whose petition was denied that year! What was the government's response?

¹The Biden administration recently announced a "supplemental cap" permitting an additional 20,000 visas for fiscal year 2022.

On June 1, 2018, OFLC announced that it would sequentially assign H-2B applications to analysts based on the calendar date **and time** on which the applications were received, based on Eastern Time, and measured to the millisecond.

Predictably, this didn't fix the problem.

Within the first five minutes of opening the semi-annual H-2B certification process on January 1, 2019, the Department's network infrastructure supporting OFLC's electronic filing system experienced more than 22,900 server login attempts.

This unprecedented volume of simultaneous system users—30 times the number of users in the previous year—ultimately caused the electronic filing system to become unresponsive and prevented almost all employers from filing H-2B applications. Although the Department was able to restore OFLC's electronic filing system by **January 7**, 2019, some employers continued to report technical difficulties with accessing the electronic filing system.

For these reasons, OFLC has concluded that changes to the procedures under which H-2B applications are assigned to NPC analysts are necessary to promote a more orderly and fair process for all employers seeking access to the H-2B visa program.

The new solution is a lottery among all applications that (i) were received within the first three days of the filing period, and (ii) propose for employment to start on the earliest permissible start date of work (April 1). Applications that don't meet both criteria are in principle valid, but in practice have no chance of being approved: last month, applications meeting both conditions accounted for 136,555 worker positions (recall that only 33,000 visas will be granted).

This post will take an employer-centric perspective, and will make several simplifying assumptions. In particular, I assume that employers care only about the *number* of H2B workers that they employ (and not the identity of these workers), that there is a large pool of potential labor available to each employer, and that application costs are small enough to be ignored. These assumptions make for a clean story, but arguably remove emphasis from a major set of stakeholders: the workers themselves! As you read, I encourage you to keep in mind that each approved or denied visa determines whether an individual person is able to earn a living (for themselves and often their family).

Current Approach: Employer Lottery

H2B visa applications are submitted by **employers**, not potential workers. Each employer's application identifies the individuals they wish to hire, and the type of work that they would be hired to do. This year, 7,875 employers submitted applications that were considered in the lottery. The USCIS processes employers' requests in a random order, giving each employer its full request until the cap is reached or all employers have been processed. Thus, employers typically get all of the visas they request, or none of them.² I call this approach an **Employer Lottery**.

The Employer Lottery is arguably fair, in that each employer has a (nearly) equal chance of having its request granted.³ However, it can lead to an outcome in which most visas go to large employers. On average, employers applying this January for visas starting April 1 requested approximately 17 visas (136,555/7,875). I don't have data for the distribution of these requests, but data from applications entered into the 2021 lottery are available. In this data, 5,404 applications were submitted for 98,022 visas: 44 employers requested a single visa, 2,193 requested fewer than 10, while 347 requested 50 or more positions. 62 requested over 100 visas, and the largest requested was for 375! If the success probability is approximately 33% (33,000 granted/98,022

²I don't know exactly what happens when the cap is reached. That is, if an employer requested 50 visas but only 30 remain, is the employer given 30 visas, 50, or 0? For this post, I assume that the answer is 30, implying that one employer might receive only part of their request. In practice, another reason that employers may get fewer visas than requested is that prospective employees are occasionally determined to be ineligible.

³Technically, employers with different request sizes have slightly different chances of being processed. However, the differences are fairly negligible: for a rigorous analysis of these differences, see the "Group Lottery" section of my paper with Carlos Bonet, Lotteries for Shared Experiences.

requested), then this large employer receives an average of 125 visas. A small employer who requests 5 visas might argue that it is unfair to deny their request while giving so many visas to large employers.

Alternative #1: Weighted Lottery

One way to address this concern would be to give employers who request more visas a lower chance of success. One simple way to do this would be to make each employer's success probability inversely proportional to their request, so that the expected number of visas allocated to each employer is the same. More formally, if our visa cap is C and employer e requests n_e visas, then the success probability of employer e is $p_e = \min(q/n_e, 1)$, where the value q satisfies $\sum_e \min(q, n_e) = C$.

In addition to ensuring that each employer is expected to get the same number of visas, this approach has another potential benefit: it might encourage employers to limit unnecessary requests. For example, suppose that an employer is considering requesting 40 visas, and anticipates a 40% probability of success. This employer might decide to reduce their request to only 20 visas, thereby doubling their chance of success to 80%. If employers believe that hiring 20 workers is more than half as good as hiring 40 workers (i.e. if they get diminishing marginal returns from each additional worker), then this could lead to a more efficient outcome.

Of course, if employers really do have diminishing marginal returns from additional workers, then perhaps the USCIS should abandon "all or nothing" approval, and instead approve only a fraction of the visas that each employer requests.

Alternative #2: Individual Lottery

This thought motivates several potential alternatives. Rather giving each employer their full request or nothing, one natural alternative is to run an **Individual Lottery**, in which each individual applicant is given a distinct lottery number. This will ensure that more employers receive at least *some* of the workers that they requested, and should lead to a more efficient allocation if employers face diminishing marginal returns. A similar approach that reduces the randomness for employers is a **Proportional Allocation**: if the total request is three times the cap, give each employer one-third of its request (some rounding required).

While seemingly natural, either of these requests would likely produce undesirable outcomes. Consider an employer who wishes to hire 10 employees and expects that only one third of its requests will be approved. Its natural response is to decide to file applications on behalf of (approximately) 30 employees! If all employers follow this logic, the result will be a chaotic spiral in which employers request ever-more visas, the visa approval rate plunges, lots of unnecessary paperwork is completed, and lots of workers get their hopes up only to later be denied. By contrast, the Employer Lottery gives no incentive for employers to inflate their demand: this will not increase their chance of being selected, and if selected, each employer will receive its full request.

Incidentally, the problem of rationing resulting in inflated demand is actually quite widespread. Think back to the beginning of the pandemic, when toilet paper started to disappear from stores. This caused a lot of people to stock up and buy far more toilet paper than usual, exacerbating the problem. In a previous post, I point out that a related issue arose with Beijing's license plate lotteries.

Alternative #3: Uniform Allocation

So an individual lottery (or proportional allocation) isn't a good idea, because it incentivizes employers to race to inflate their demand. Is there an alternative that allows employers to receive a fraction of their request, while eliminating incentives to request more workers than needed?

This question was studied in the paper Capacity Choice and Allocation: Strategic Behavior and Supply Chain Performance by Gerard Cachon and Marty Lariviere. One simple approach that they discuss – which I will call a **Uniform Allocation** – works as follows:

Allocate in a series of “rounds.” In each round, allocate one additional visa to each employer which has not yet received its full request. Continue until the cap is reached, or all employers have received their full request. (If there are not enough remaining visas to complete a round, a priority order can be used to determine which employers get the last visas.)

Another way of thinking about a Uniform Allocation is that it starts by dividing visas evenly among all employers: if the cap is 1000 visas and 50 employers applied, each one is offered 20 visas. If every employer wanted at least 20 visas, the allocation is finalized. Most likely, however, some employers requested fewer than 20 visas. In this case, any visas that they didn’t claim are redistributed evenly among the remaining employers. This process continues until all visas are claimed or all employers receive their full request.

In fact, the Uniform Allocation and the Employer Lottery are both special cases of a more general family of procedures.

A **sequential claim** mechanism is characterized by a finite sequence of employer ID numbers (each employer ID can appear many times in this sequence). This sequence is processed in order, with the employer whose ID comes next given the opportunity to claim one additional visa. This process continues until the cap is reached or the sequence is fully processed. Because it would be impractical to run this procedure in real time, we consider a “direct” implementation, in which each employer requests a number of visas, and each time it receives an offer, it accepts so long as it has not yet reached this number.

Note that with a cap of C , the Employer Lottery can be thought of as a sequential claim mechanism in which each employer ID is in C consecutive positions, while the Uniform Allocation is a sequential claim mechanism that lists each employer once before listing any employer twice.

I claim that in any sequential claim mechanism, employers do not have an incentive to inflate their demand. To make this claim formal, I describe the utility of each employer with a function $u : \mathbb{N} \rightarrow \mathbb{R}$, with $u(k)$ interpreted as the utility of receiving k H2B visas. Say that an employer has **unimodal preferences** if its utility function is unimodal. This means that there is an optimal number of visas that it would like to receive: until it reaches this number, each additional visa is beneficial, but beyond this number, additional visas are undesirable.

Theorem In any sequential claim mechanism, any employer with unimodal preferences has a dominant strategy of reporting their optimal number of visas.

Note that this result allows any sequence of offers, so long as this sequence does **not** depend on the number of visas requested by each employer.

Comparing the Employer Lottery to Uniform Allocation

One important consequence of moving from an employer lottery to a uniform allocation would be a shift of visas towards small employers. Let’s consider a simple example. There are 100 small employers who request two visas, 100 medium employers who request 10 visas, and 100 large employers who request 50 visas. Thus, a total of 6200 visas are requested. However, the total number of visas awarded is capped at 1240, which is 20% of the total request.

- With an employer lottery, we expect to select 20 small employers, 20 medium employers, and 20 large employers. In total, 40 of our 1240 visas (3.2%) go to small employers, with 200 (16.1%) going to medium employers and 1000 (80.6%) going to large employers.
- With a uniform allocation, we have enough visas for each employer to get (at least) 4 visas. Thus, small employers are given their complete request of 2 visas. Redistributing the visas that small employers did not claim allows us to give each medium and large employer a fifth visas. This accounts for 1200 visas: the remaining 40 can be awarded randomly to medium and large employers, so that some are approved for a sixth visa. In total, 200 visas go to small employers (16.1% of the total), 520 go to medium employers (41.9% of the total), and 520 go to large employers (41.9% of the total).

As you can see, there is quite a dramatic difference between these two! Using the aforementioned data the 2021 lottery, we see that just over 40% of applications are submitted by small employers (< 10 visas requested),

while 6.4% are submitted by large employers (50+ visas requested). The current employer lottery is expected to award only 3999 visas to “small employers” (12% of the total), and 9,694 visas to large employers (29% of the total). That is, under the current system, large employers collectively receive nearly 2.5 times as many visas as small ones. A uniform allocation would grant each employer who requested 6 or fewer visas their full allotment. Under this system, small employers would receive 11,124 visas (34% of the total). This is nearly three times their allotment under the employer lottery! Meanwhile, large employers would see their collective allotment decline by 75%, to 2,366 visas (7% of the total).

This proposed change would also be likely to have ripple effects on the behavior of potential H2B recipients. Currently, their probability of receiving a visa is roughly the same, regardless of which employer sponsors them. Under a uniform allocation, workers sponsored by large employers would be much less likely to be approved than those sponsored by small employers. This could cause workers to be reluctant to be sponsored by large employers, adding yet another reason that moving to a uniform allocation would favor small employers over large.

A Hybrid Approach

One concern with a uniform allocation is that it assumes employers are willing to hire a fraction of their requested total. An employer who requests 100 visas might have planned to use these workers for a large project. If told that it can hire only 12 workers, it might genuinely prefer to cancel the project and hire nobody.

One possible solution is to adopt a hybrid approach which combines alternatives 1 and 3. As part of their application, employers would be asked whether they would prefer to receive their share of visas with certainty, or to be entered into an all-or-nothing lottery to determine whether their requests will be approved. Employers who indicated a preference for partial allocation would receive their outcome from a uniform allocation. Employers who indicated a preference for a lottery would have a success probability set so that the expected number of visas approved was equal to their outcome from a uniform allocation. That is, employers with larger requests who opt for the lottery would have a lower success probability, but would receive more visas when successful.

Of course, there are many other possible hybrid schemes. In principle, employers could be asked to specify a preferred distribution of outcomes, subject to a constraint on the mean of this distribution. It seems much more practical, however, to ask a simple yes/no question about whether they would prefer a partial allocation or an all-or-nothing lottery.

Employers Submitting Multiple Applications

So far, I have assumed that each employer can submit only a single application. In the current employer lottery, it is clear that employers could increase their chances by submitting multiple applications, which collectively request more visas than are truly needed. I had assumed that multiple applications from the same company are not allowed, but the data from 2021 seems to suggest otherwise.

Among applications entered into the 2021 lottery, over 150 list the employer address as 980 Jolly Road, Suite 300, Blue Bell PA, the headquarters of Brightview. Each of these 150 applications technically has a different employer name. For example, one is “BrightView Landscape Development, Inc.- Denver, CO”, and another is BrightView Landscape Development, Inc. - Phoenix, AZ”. So this looks like a national company with many different locations, submitting separate requests for each location. Maybe that’s reasonable. Dig deeper, however, and we see the following:

- BrightView Landscape Development, Inc.- Denver, CO: 28 Landscape Laborers (**Granted**)
- BrightView Landscape Services, Inc. - Denver, CO (North): 50 Landscape Laborers (**Withdrawn**)
- BrightView Landscape Services, Inc.- Denver, CO (South): 220 Landscape Laborers (**Granted**)
- BrightView Landscape Services, Inc. - Denver, CO (West): 60 Landscape Laborers (**Withdrawn**)
- BrightView Landscape Services, Inc. - Denver, CO (East): 50 Landscape Laborers (**Withdrawn**)

This invites some further questions. I see that the “decision dates” for the withdrawn requests above are all January 21, 2021, while the decision dates for granted requests are February 11 and February 22. Perhaps the department of labor flagged these as duplicates and asked Brightview to withdraw some of their applications. Some evidence for this is that this year, we see an application from “BrightView Landscape Services, Inc. – Denver Branches” (rather than separate applications for North, South, West, and East). This January 7th, the department of labor published a list of applicants, alongside the lottery group into which they had been placed (A-G, with A being highest priority). “Brightview Landscape Services, Inc. - Menlo Park” was placed into group A, while “Brightview Landscape Services, Inc.- San Jose” was placed into Group E. These locations are quite close to each other, and employees could easily drive between them. Even if the department of labor flagged these as duplicate applications and asked Brightview to withdraw one, Brightview now knows which to withdraw.

While I don’t fully understand what is happening, it certainly seems as though companies are able to increase their chances by applying multiple times. If this is the case, it suggests that the current Employer Lottery is not so fair, and also not very stable (more employers are likely to catch on to this loophole over time). A uniform allocation would offer perhaps even more of an incentive to submit multiple applications. Interestingly, the Individual Lottery and Proportional Allocation – both of which incentivize employers to inflate their demand – do not incentivize the submission of multiple applications. Meanwhile, a first come first served allocation (assuming it can be feasibly implemented) does not incentivize either inflated demand or multiple corporate identities.